



UC San Diego

JACOBS SCHOOL OF ENGINEERING
Electrical and Computer Engineering



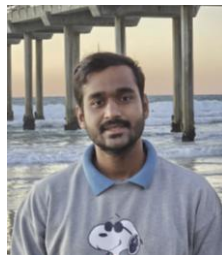
SenSync: Real-Time and Accurate Passive Sensing



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Gupta

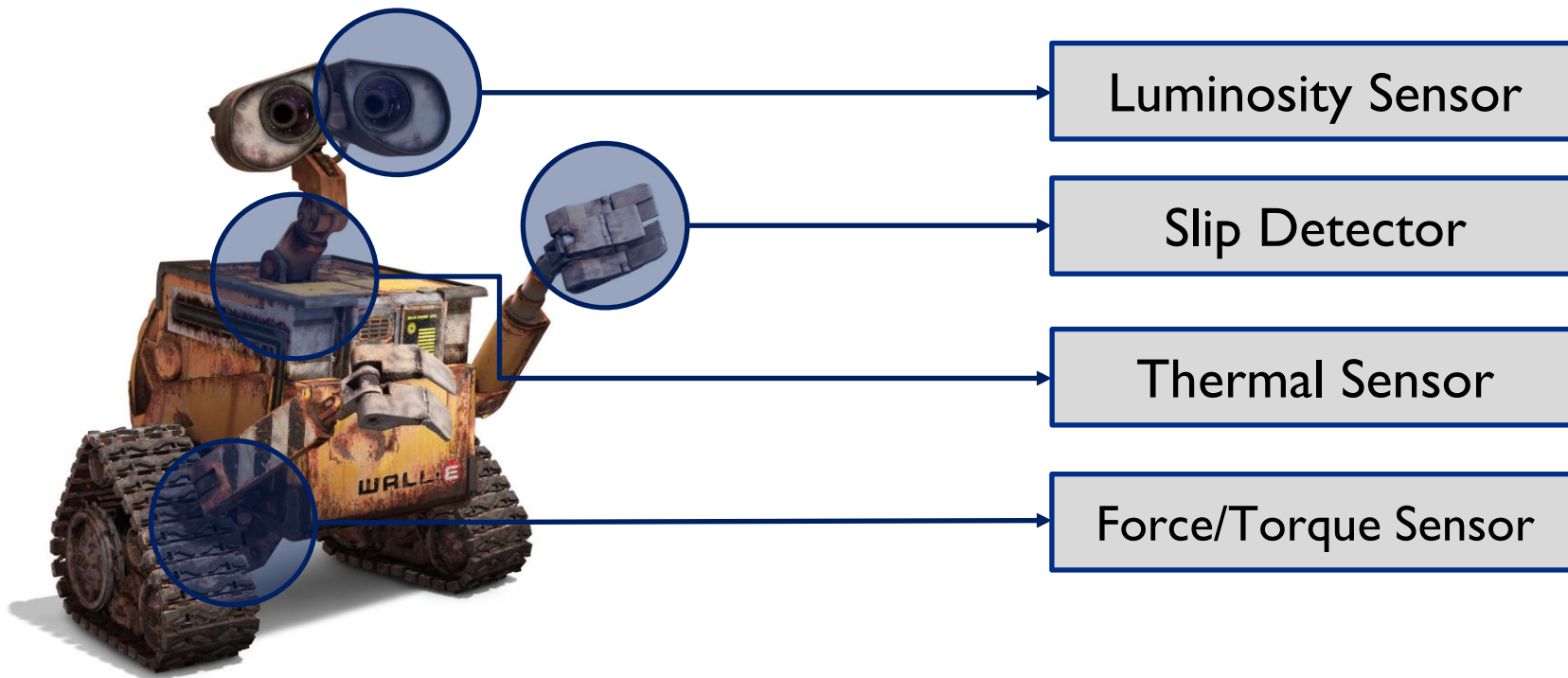


Harine
Govindarajan

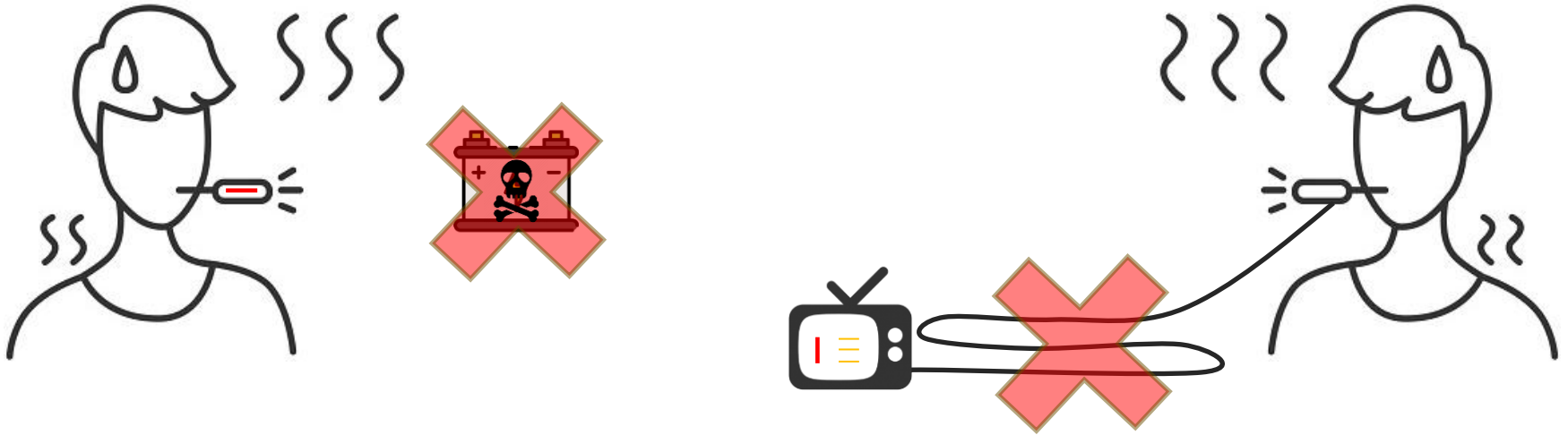


Prof. Dinesh
Bharadia

Sensing the World, One Byte at a Time

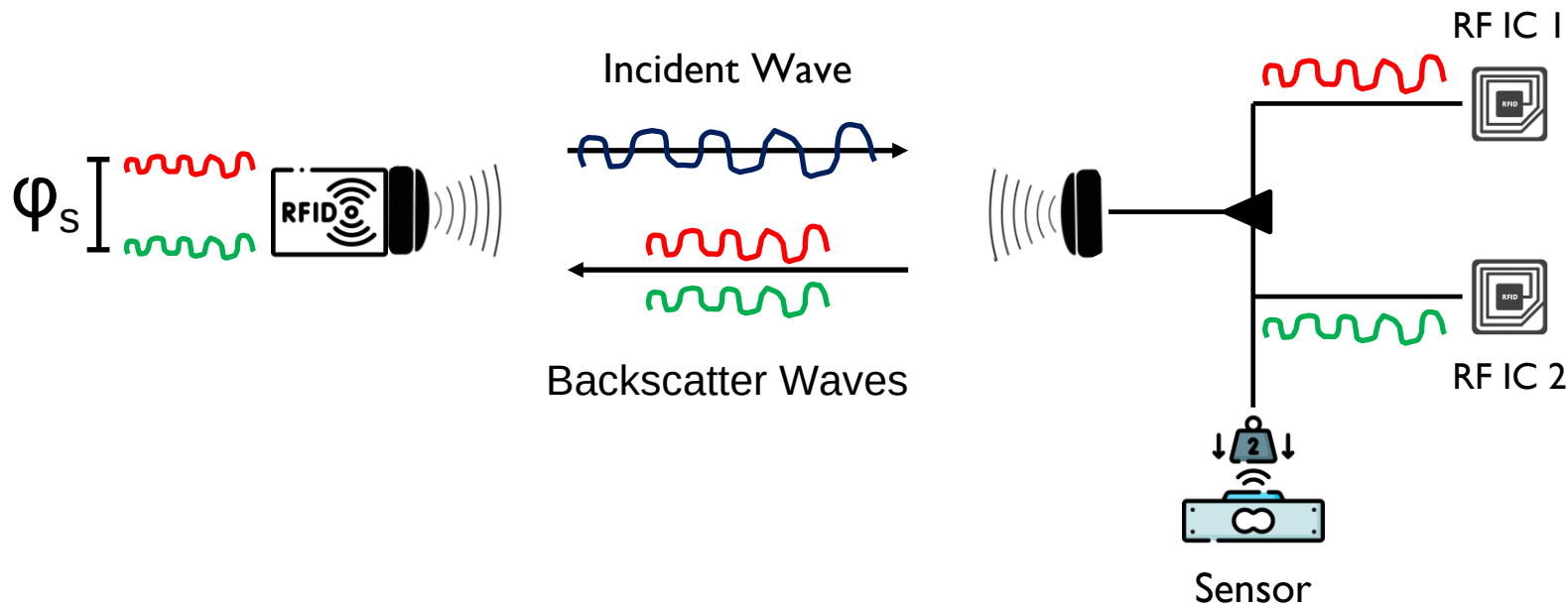


Not so Ubiquitous – The Ball and Chain Problem



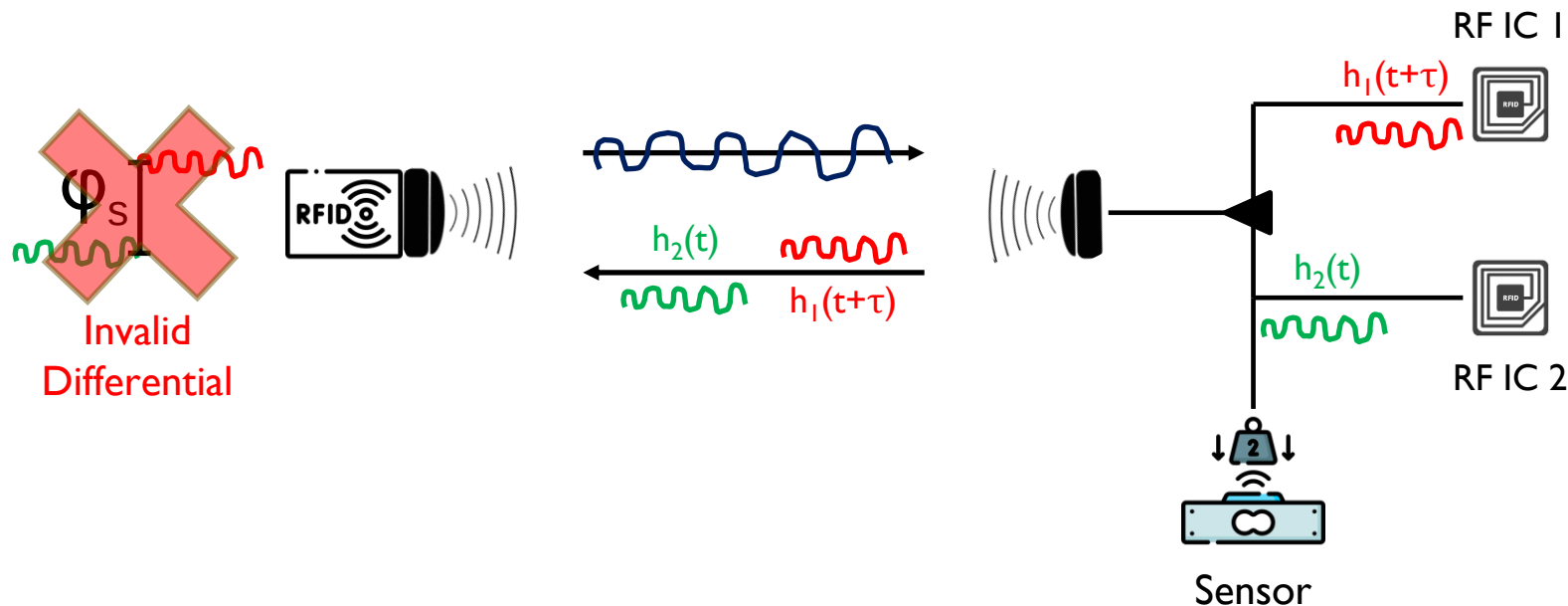
Wires and Batteries are the shackles for Ubiquitous Sensing

Unplugging the Future: RFID Sensor Networks



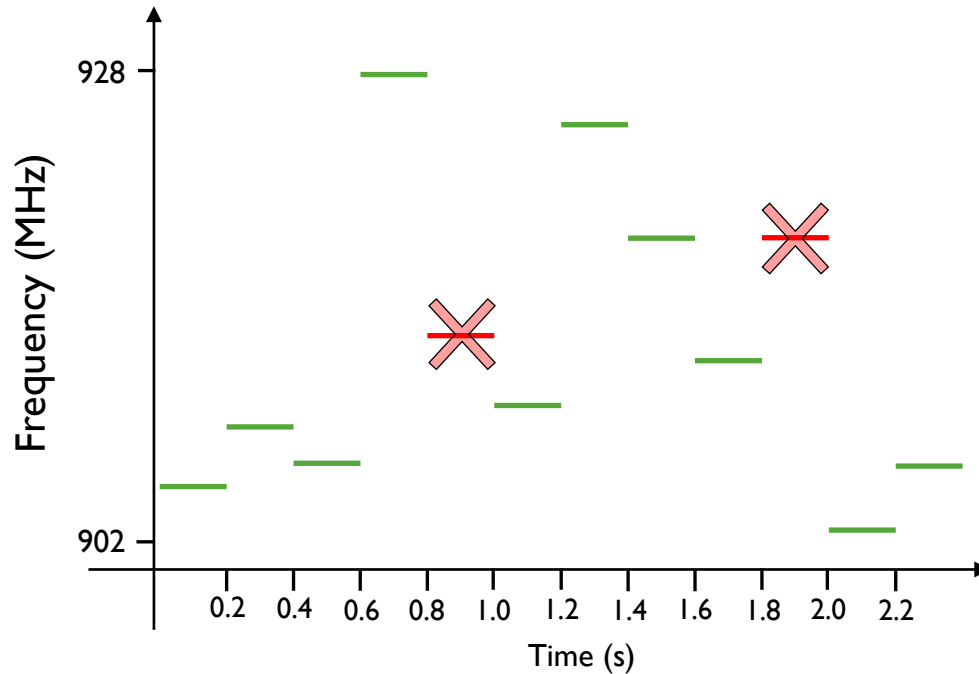
Beyond the Noise: Unlocking Accuracy with Differential Sensing^{1,2}

Is Differential the Answer... Or Just Another Layer?



Sequential Reads: The Devil in the Data's Details

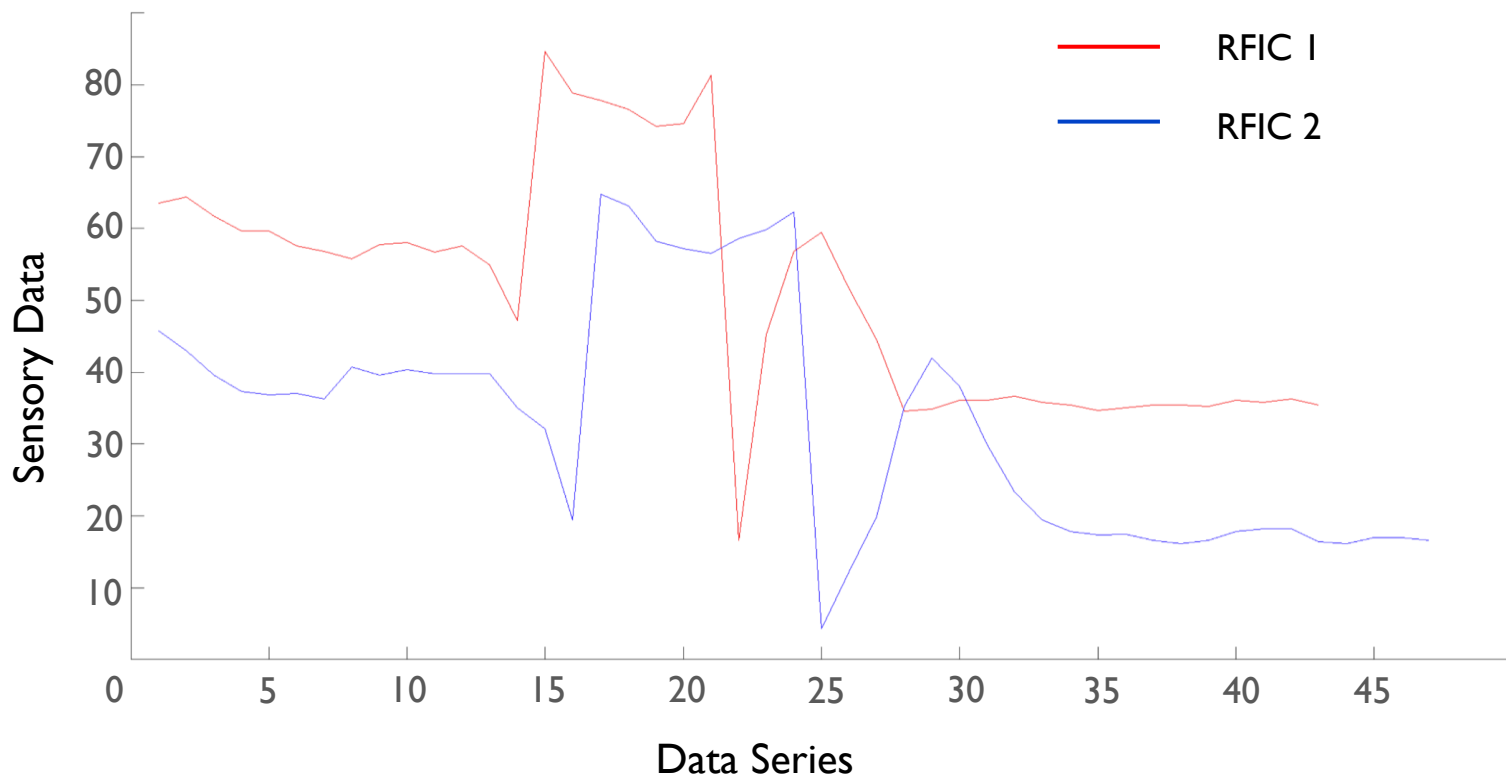
Two Signals Hopped to a Bar...



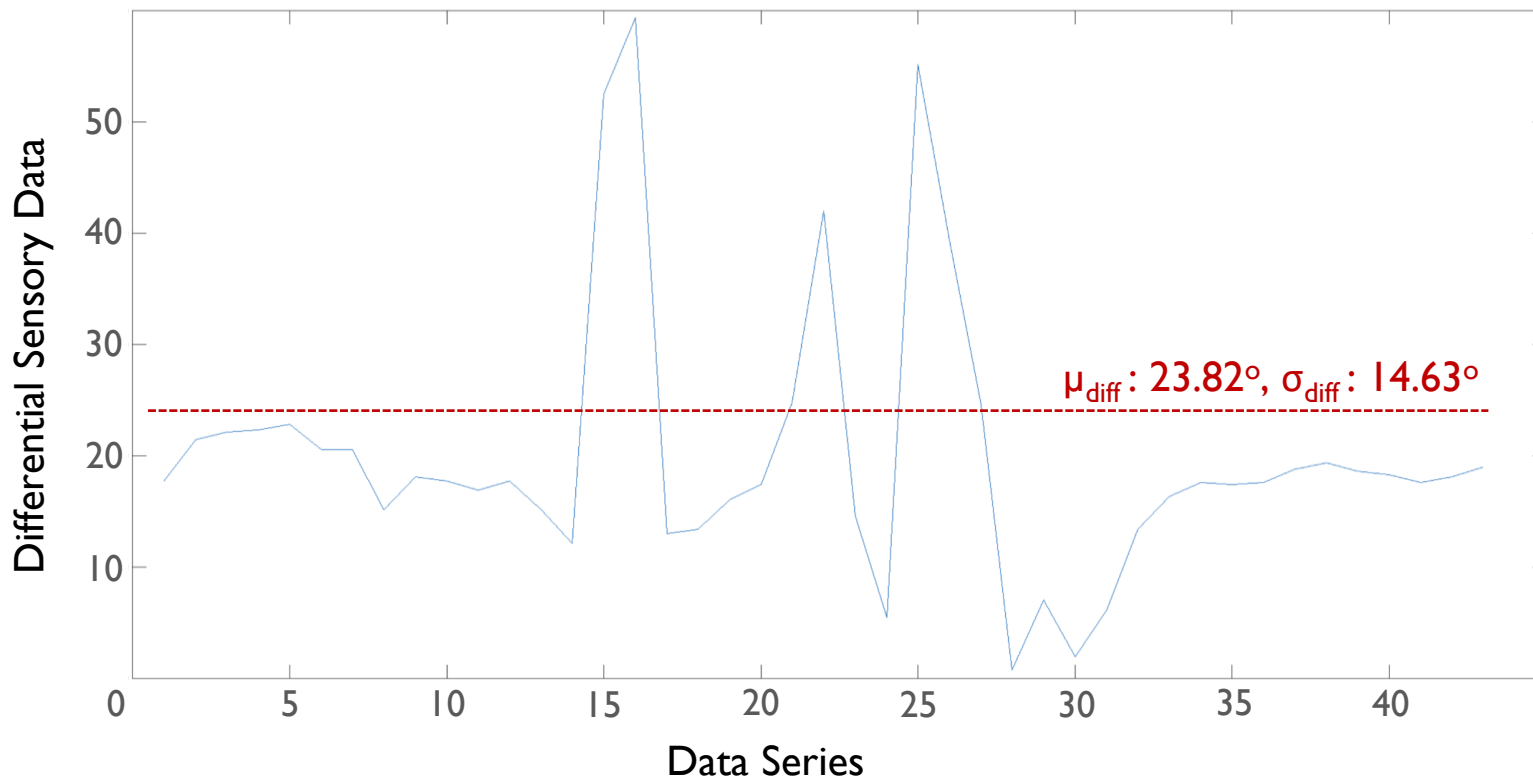
- FCC requires the reader to hop frequencies to prevent congestion which happens every **200ms**
- Readers have an internal caveat which forces arbitrary phase jumps of π **15%** of the times
- This makes phase a **volatile** and **unreliable** metric for direct sensory measurement

Phase Detection fails when the Carrier Jumps

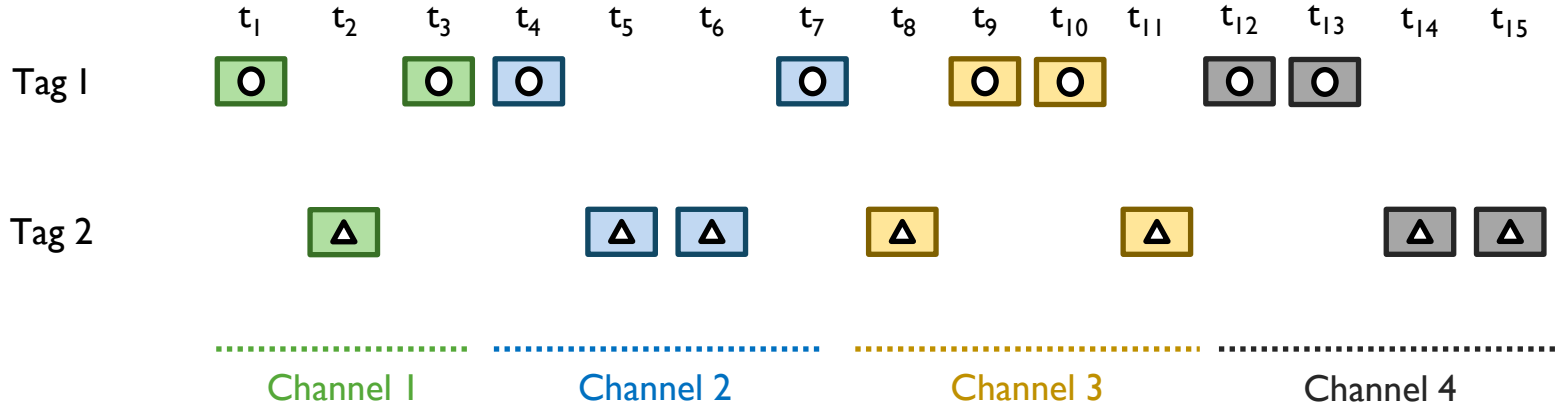
How do Things Stack up in Reality...



Like a Real Mess...



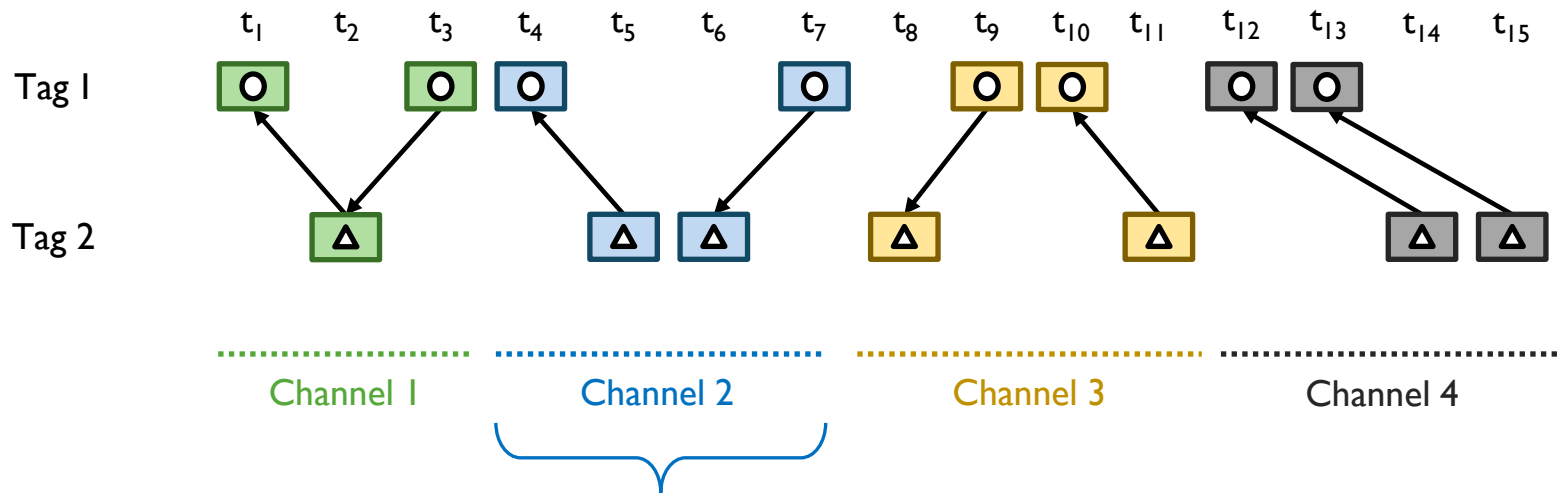
Solving the Data Puzzle



$$\Phi_1 = \{ \Phi_1^c(t_n) \mid c \in \{1, 2, \dots, C\}, n \in \{1, 2, \dots, N\} \}$$

$$\Phi_2 = \{ \Phi_2^c(t_m + \tau_{c,m}) \mid c \in \{1, 2, \dots, C\}, m \in \{1, 2, \dots, M\} \}$$

Secret Sauce: Dynamic Time Warping

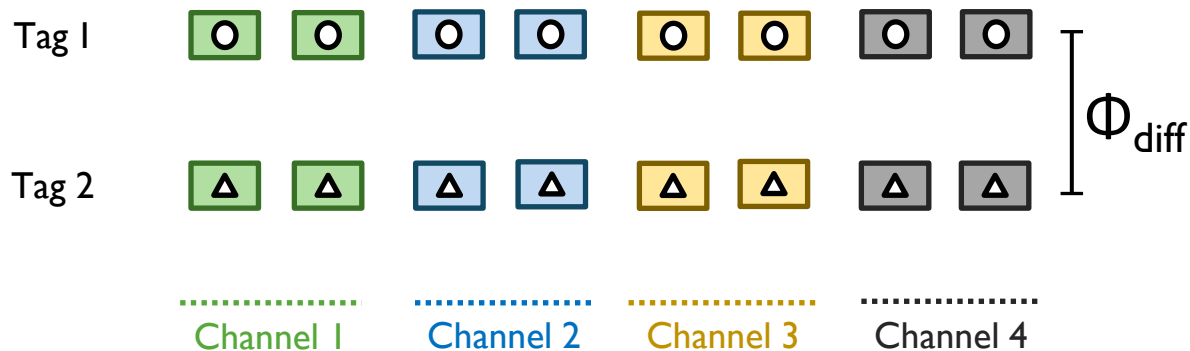


$$\Phi_1 = \{ \Phi_1(t_i) \} \forall t_i \in W$$

$$\Phi_2 = \{ \Phi_2(t_i) \} \forall t_i \in W$$

... W = Warping Path from DTW

Unlocking Similarity: Art of Alignment



Empirical data shows that 15% of the frequency hops are corrupted

We chose 4 channels for DTW frame since there is a 90% chance that at least 3 channels are accurate

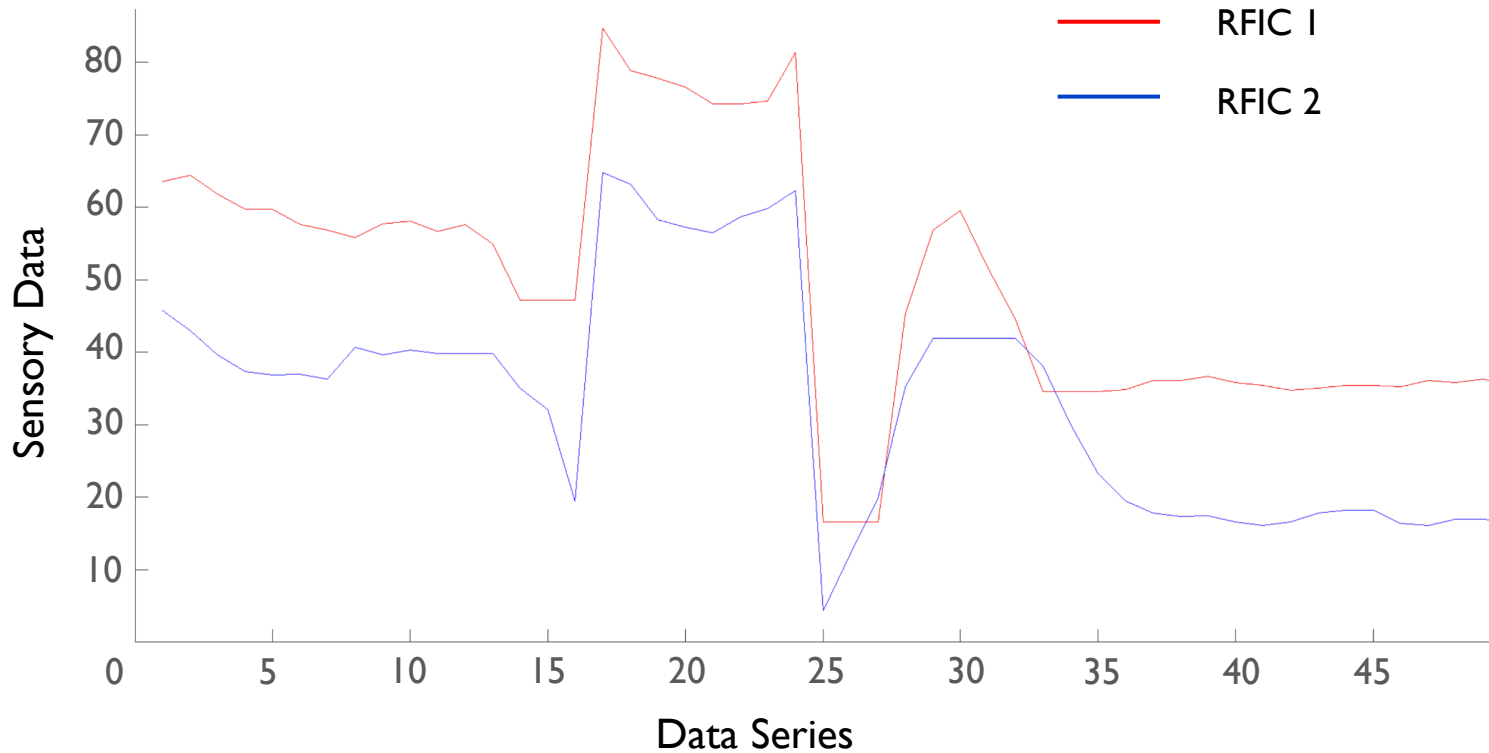
$$\Phi_{diff}^c(t_i) = | \Phi_1^c(t_i) - \Phi_2^c(t_i) | \forall t_i \in W, c \in \{1, 2, 3, 4\}$$

$$\Phi_{diff} = (\overline{\Phi_{diff}^1(t_i)} + \overline{\Phi_{diff}^2(t_i)} + \overline{\Phi_{diff}^3(t_i)} + \overline{\Phi_{diff}^4(t_i)}) / \sum_1^N i$$

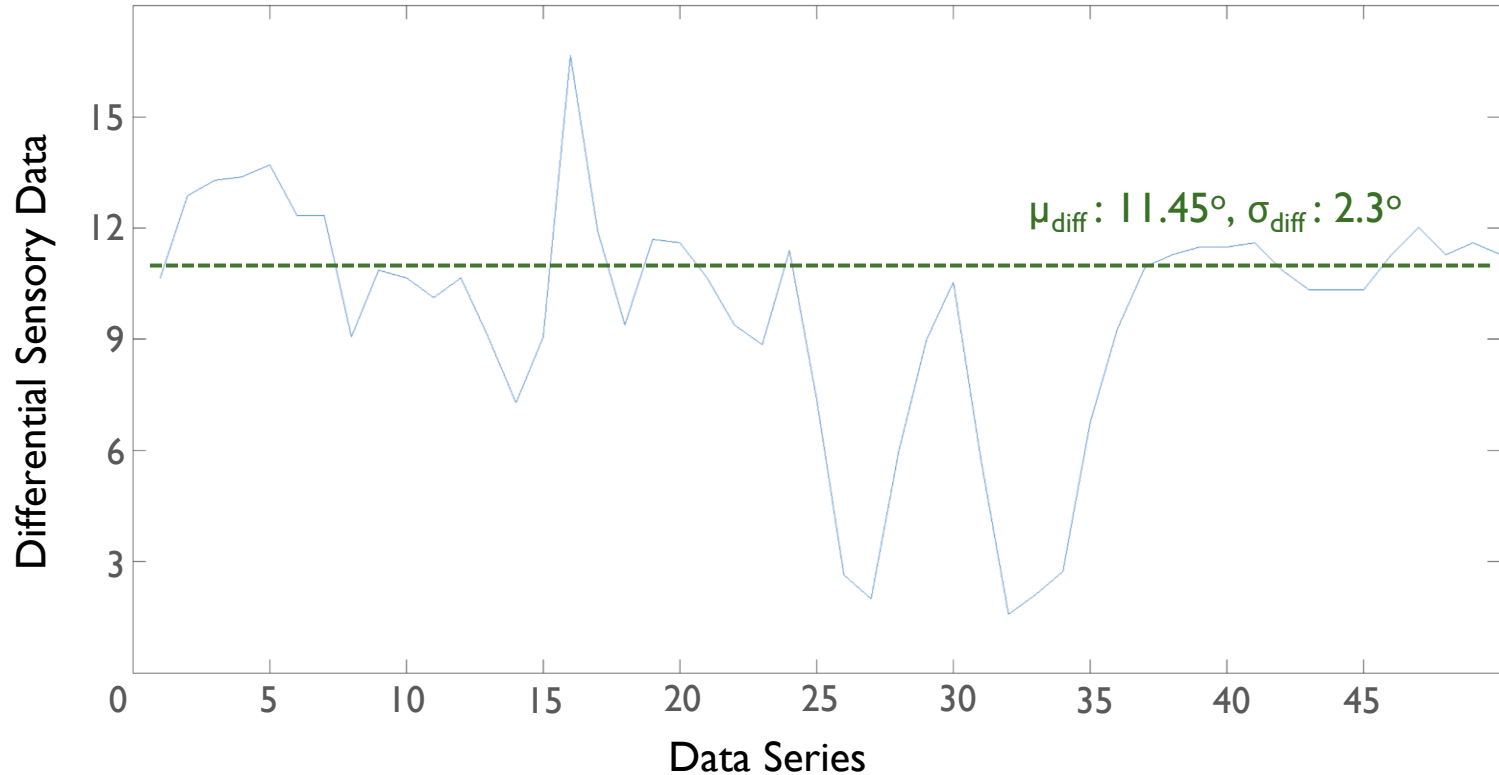
Each channel is 200ms, which gives us a resolution latency of 0.8s

Resolves chaotic RFID signals into clear outcomes **5x** faster than State of Art

Reality Realigned: Temporal Distortions Addressed

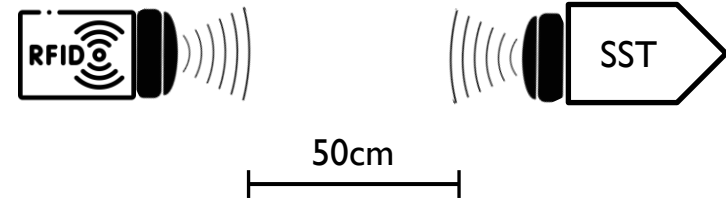
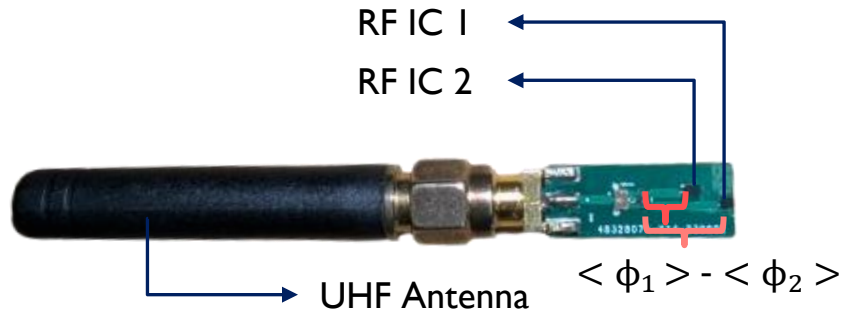


Cleaning up the Mess...



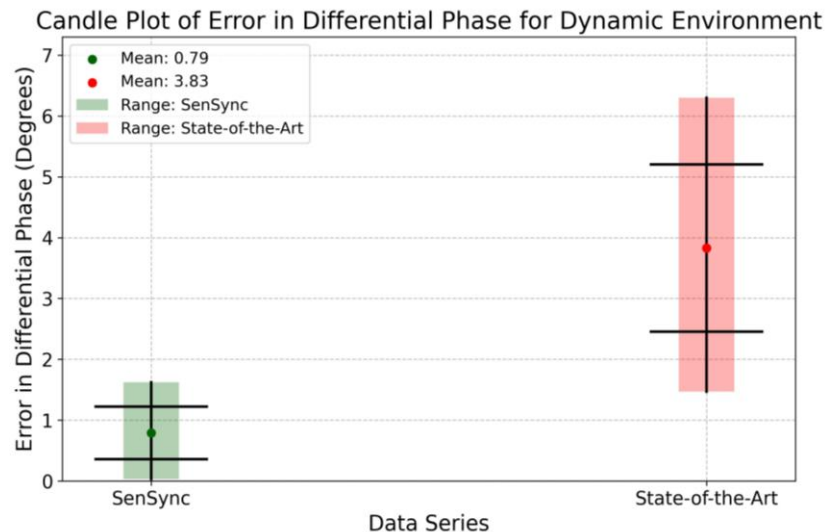
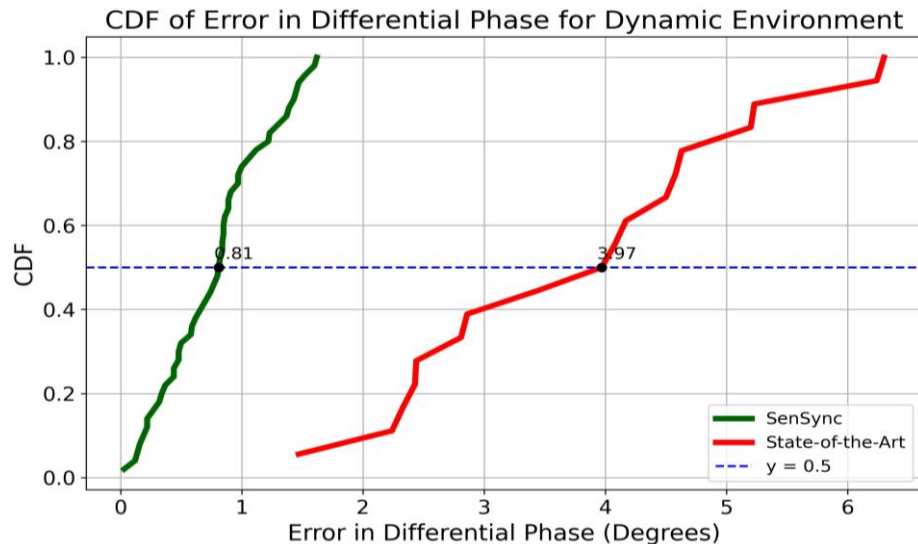
The Proof of the Pudding...

Simulatory Stubbed Tag



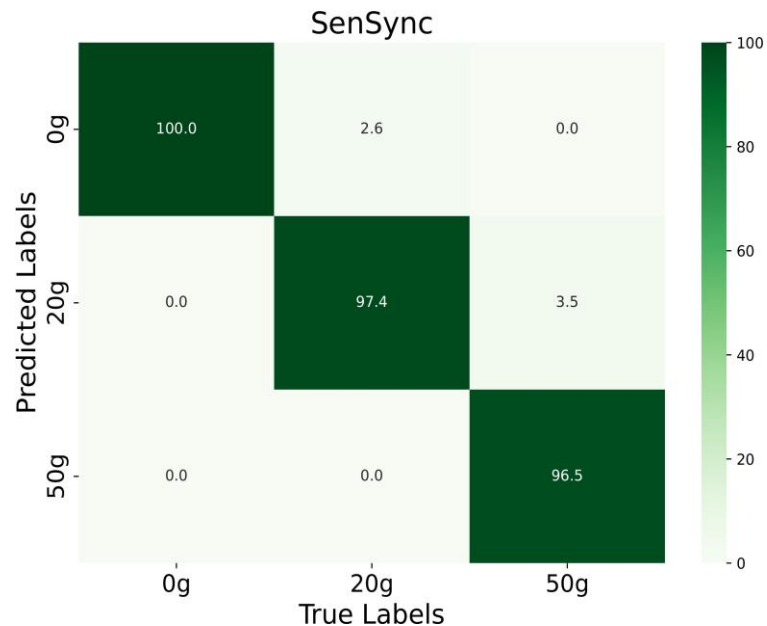
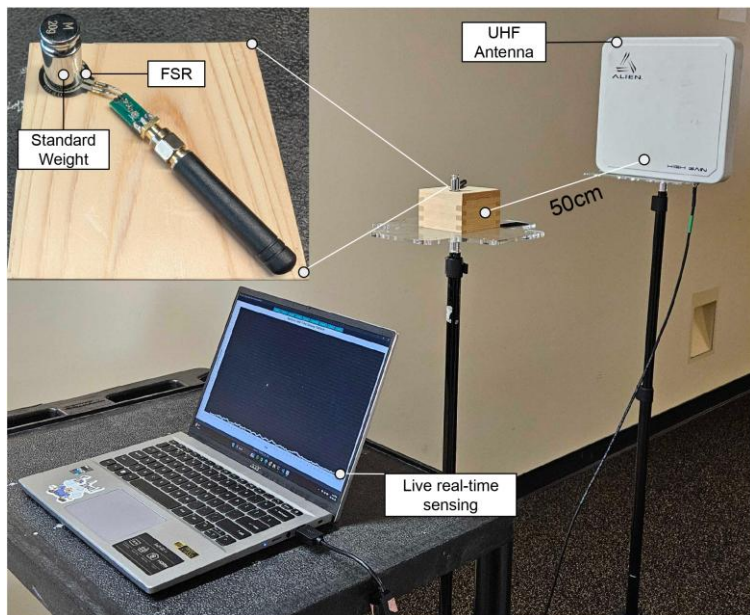
$$\Delta\varphi = \Delta\text{Electrical Length} = 5.95^\circ \longrightarrow \text{Round-Trip } \Delta\varphi = 11.9^\circ$$

... is in the Eating



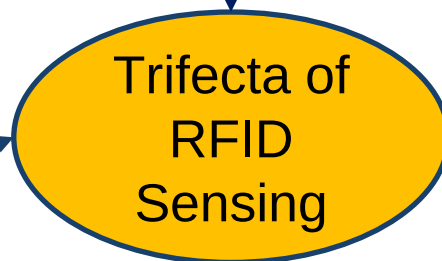
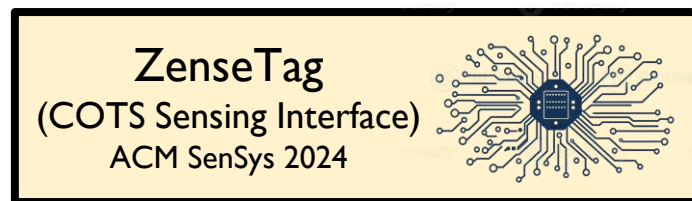
When Less is More: SenSync is **4x** better than State-of-the-Art sensing systems

Closing the Loop: Real World evaluation



From Lab to Life: **97%** classification accuracy when measuring weights

In Sync with the Senses



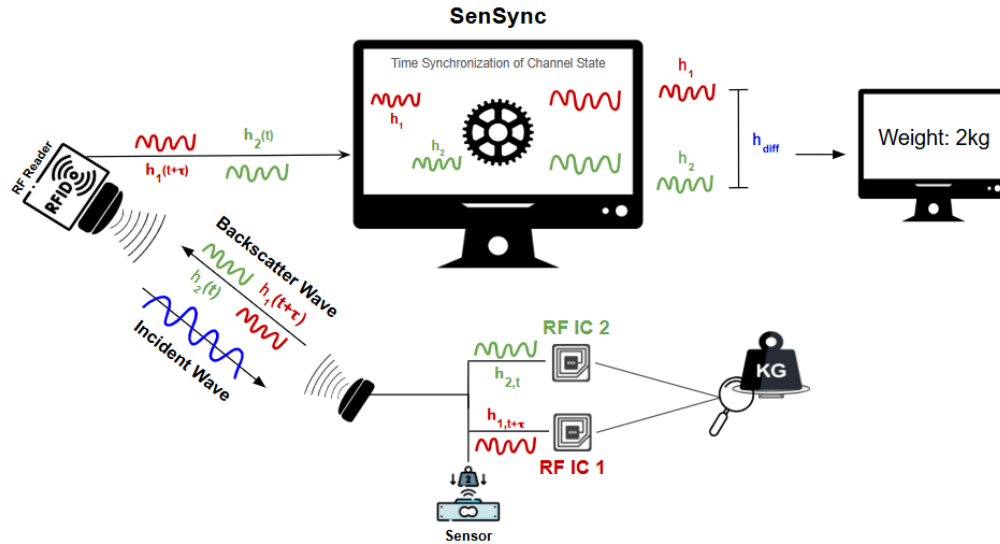
Forcesticker
(Miniaturized Force Sensor)
ACM IMWUT 2023



SenSync
(Real-Time Algorithm)
IEEE RFID 2025



In a Nutshell



SenSync achieves -

- **5x** better sensing latency
- **4x** better at resolving stimuli
- **97%** weight detection accuracy

Questions ?

Check our
work out at:

